

TIC 233459038

Eclipsing binary candidate dossier - Star Vetter

PERIOD ALIAS

BLS EPHEMERIS

Period 25.338439 d (ALIASED, see below)
-> corrected period: 50.676878 d
Epoch (t0) 1728.5852 BTJD
Duration 0.1700 d
Depth 8.315 % (83,146 ppm)
SNR 236.6

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

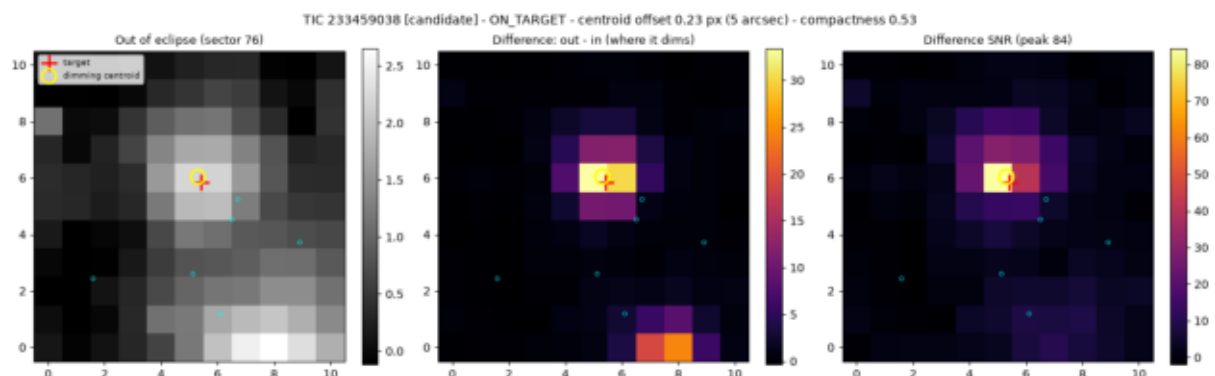
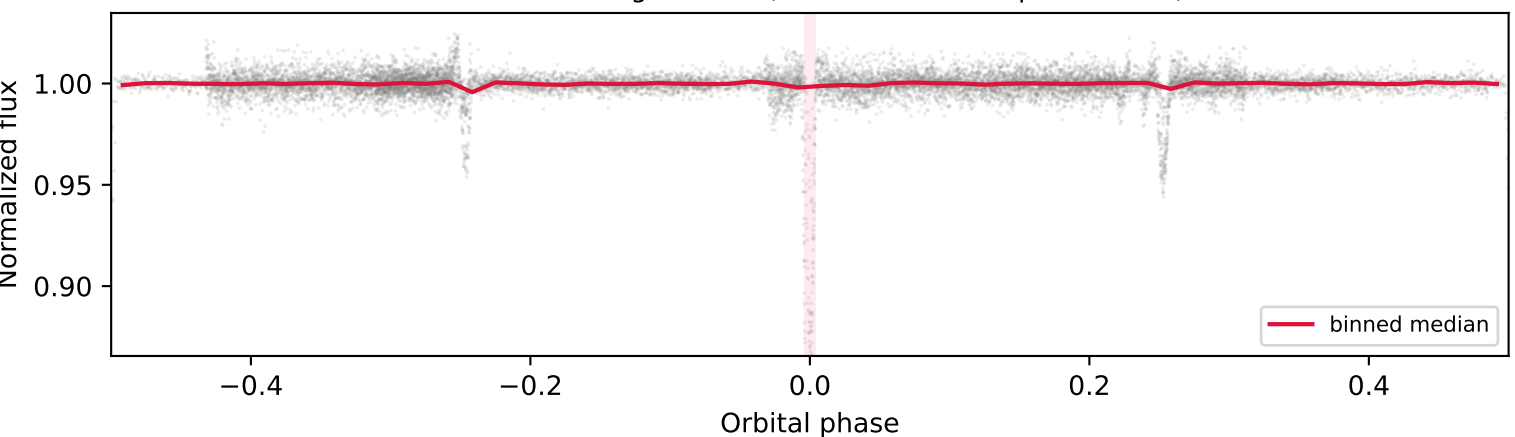
Eclipses observed 6 (gate ≥ 3)
Robust depth ratio 1.37 (gate ≥ 0.50)
Robust depth sigma 274.3 (gate ≥ 7.0)
Odd/even z-score 89.17 (ALIAS FLAGGED)
Secondary eclipse detected, phase 0.26, 8.3 sigma

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 76 light-curve dip in this sector: 169.1 sigma t0 refined by 0.0000 phase
Difference-image peak SNR 84.46 (gate ≥ 5.0) Compactness 0.53 (gate ≥ 0.50)
Dimming centroid offset 0.23 px (5 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: none within the cutout

Phase-folded light curve (shaded = BLS eclipse window)



Caveat (PERIOD ALIAS): Odd/even eclipse-depth z-score is 89.17 (gate > 3.0): alternating eclipses differ significantly in depth, the signature of a true period double what BLS found. Pixel-check confirmation and eclipse timing are unaffected - report the corrected period, 50.676878 d, in any write-up.